

CONTACT	Georgia Institute of Technology Sacramento, California, United States	<i>E-mail:</i> ethan.villalovoz@gatech.edu <i>Links:</i> Website , LinkedIn , Google Scholar
RESEARCH INTERESTS	Embodied intelligence, robot learning, and human–robot interaction.	
EDUCATION	Georgia Institute of Technology <i>M.S. in Computer Science (GPA: 4.0/4.0)</i>	Jan 2026 – May 2028
	Washington State University <i>B.S. in Computer Science (GPA: 3.94/4.0)</i>	Aug 2021 – May 2025
RESEARCH EXPERIENCE	Washington State University , Pullman, WA <i>Undergraduate Research Assistant</i> Collaborated with Profs. Janardhan Rao Doppa and Haipeng Cai on <i>Bayesian prompt optimization</i> to improve the correctness of LLM-generated code. Implemented evaluation pipelines and reproducible benchmarking experiments.	Jan 2024 – May 2025
	Carnegie Mellon University , Pittsburgh, PA <i>Robotics Institute Summer Scholar</i> Collaborated with Profs. Henny Admoni and Reid Simmons on <i>robot learning from human feedback</i> . Developed a Bayesian reward-learning system that combines state corrections with clarification questions to infer human preferences in simulation.	Jun 2024 – Aug 2024
	Oregon State University , Corvallis, OR <i>NSF REU Fellow</i> Collaborated with Prof. Heather Knight on <i>expressive multi-robot motion</i> and its influence on human perception. Designed geometric motion primitives and built Python tools for configuring robot formations. Published at IROS 2023.	Jun 2022 – Aug 2022
PROFESSIONAL EXPERIENCE	Microsoft , Redmond, WA <i>Software Engineer Intern</i> Worked with Karunakar Vecham on <i>AI agents for enterprise financial workflows</i> . Built a C#/.NET agent prototype integrating enterprise APIs and evaluated its tool use, cost, and reliability. Improved performance and remediated security vulnerabilities in production applications.	May 2026 – Jul 2026
	Google , Sunnyvale, CA <i>Software Engineering Intern (STEP)</i> Worked with Arun Tej Chennadi and Paul Valdez on <i>scalable database analytics</i> . Developed C++/SQL jobs and dashboards to automate reporting of queue metrics. Optimized data sampling to collect statistics across the full dataset.	May 2023 – Aug 2023
TEACHING EXPERIENCE	Georgia Institute of Technology , Atlanta, GA <i>Graduate Assistant</i> TA for <i>CS 8803 O27: Computer Graphics in the AI Era</i> w/ Prof. Bo Zhu . Covering NeRF, 3D Gaussian splatting, differentiable rendering, and world models. [Site]	Fall 2026
	Washington State University , Pullman, WA <i>Undergraduate Teaching Assistant</i> TA for <i>CPT S 315: Introduction to Data Mining</i> w/ Prof. Honghao Wei . Covered data preprocessing, frequent pattern mining, recommendation systems, classification, and clustering. Gave guest lectures on practical machine learning and Python libraries for course projects.	Spring 2025

Washington State University (continued)

Undergraduate Teaching Assistant

Fall 2024

TA for *CPT S 350: Design and Analysis of Algorithms* w/ Prof. [Zhe Dang](#). Covered divide and conquer, dynamic programming, greedy algorithms, graph algorithms, and complexity analysis. Oversaw the course's Python graph-search project.

Undergraduate Teaching Assistant

Fall 2023

TA for *CPT S 355: Programming Language Design* w/ Prof. [Sakire Arslan Ay](#). Covered programming paradigms, type checking, scoping, closures, and interpreter design. Guided students through programming assignments in Haskell and Python.

Undergraduate Teaching Assistant

Fall 2022

TA for *CPT S 121: Program Design and Development in C* w/ Prof. [Andrew O'Fallon](#). Covered control flow, functions, arrays, pointers, recursion, and debugging. Led programming labs and held office hours.

HONORS

Generation Google Scholarship

2023

Hispanic Scholarship Fund Scholar

2022, 2023

CONFERENCE
PUBLICATIONS

[C1] **Social Triangles and Aggressive Lines: Multi-Robot Formations Impact Navigation and Approach.**

A. Bacula, [E. Villalovoz](#), D. Flynn, A. Mehta, H. Knight.

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023. [\[PDF\]](#)

PRE-PRINTS

[P1] **An Exploratory Study of Bayesian Prompt Optimization for Test-Driven Code Generation with Large Language Models.**

S. Tomar, A. Deshwal, [E. Villalovoz](#), M. Fazzini, H. Cai, J.R. Doppa.

arXiv:2512.15076, 2025.

[\[PDF\]](#)